

Energy-aware Model Predictive Control with Obstacle Avoidance

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Abstract—This work proposes a single-layer finite-horizon optimal control strategy to solve the autonomous navigation problem while accounting for energy efficiency and providing obstacle avoidance feature in cluttered environments with unknown obstacles. Considering the rate capacity effect of electric batteries, the nonlinear state-of-charge behavior is described and included in the optimal control problem to achieve energy-awareness. Besides, artificial potential fields are considered to obtain obstacle avoidance capabilities. The control problem is formulated inspired by the tracking model predictive control framework, and it considers the central idea of including artificial variables into the control problem to obtain a closed-loop system with an enlarged domain of attraction and with feasibility insurance. Finally, numerical results in a case study considering a quadrotor UAV are provided to corroborate the proposed strategy.

I. INTRODUCTION

Systems able to perform autonomous navigation are expected to play an important role in the upcoming years in the most distinct types of applications. Thus, research solutions for problems related to how these systems can navigate safely and efficiently in a given environment are imperative. Particularly, being able to avoid collisions during navigation without losing sight of energy efficiency are important features required to such systems. In fact, the ability to handle unexpected situations that could present a risk for the motion system and its surroundings is necessary to achieve fully autonomous systems. Moreover, the problem of endurance of systems powered by batteries has prevented them to be widely used and, therefore, it poses an important issue related to the advance of this kind of systems.

Although those two problems could be handled decoupled from the motion control problem by considering planning and power management strategies, seeking a solution through unified approaches could prevent suboptimal responses and performance deterioration. Among the different types of strategies capable of dealing with both problems in a unified fashion, optimization-based approaches can provide ingredients, such as cost functional and constraints, that could easily allow the development of novel strategies gathering

distinct objectives together. Besides, among the optimal control techniques, Model Predictive Control (MPC) strategies have stood out due to their ability to handle in a simple way multivariate underactuated systems, ensuring input-to-output stability and internal stability [1]. Moreover, their prediction-based nature allows obtaining a control system capable of providing smoother signals with less overshoots, which, for autonomous navigation, results in a better tracking performance with smaller control effort.

The problem of endurance during autonomous navigation is related to how the efficiency of the motion system power source can be maximized. Particularly, systems storing energy in batteries have taken advantage of the great effort being made in material engineering and chemistry to improve battery technology as the world drives towards renewable and sustainable energy sources [2]. Also, some researches achieve enhanced endurance by considering electric batteries working with different power supplies for recharge, for instance, photovoltaic and fuel cells [3], [4]. Although those efforts are essential and have led to the increasing endurance of autonomous systems powered by batteries, it is also possible to contribute by designing energy-aware control and planning algorithms to perform autonomous navigation while maximizing energy efficiency. With that purpose, considering requirements such as the energy consumption of the actuators, the minimum energy consumption path, the charging and discharging battery dynamics, the battery best-operating range, and, even, the conditions to extend the battery lifespan, are important contributions to the problem.

From the energy-awareness perspective, some works consider the design of algorithms to define energy-efficient paths [5], [6] while others consider the definition of energy optimal control [7], [8]. Particularly, [7] consider the battery dynamics in a simplified way through the rate capacity effect and define operational constraints to it. Besides, [9] and [10] show the possibility to use the economic MPC framework for that problem. In fact, knowing the battery dynamics, either based on a capacitor-resistor equivalent circuit [11] or based on the electrochemical processes [12], it is possible to define additional costs to bring the required motion control and energy efficiency problems together.

Regarding obstacle avoidance, many approaches for planning and control can be found in the literature [13]. In particular, the planning problem of finding collision-free paths towards a given configuration has been approached in different ways in the last decades, going from classical to heuristic-based techniques [14]. There are works in the literature already dealing with both problems through non-decoupled schemes considering optimal control [15], [16];

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however, most of them are based on the design of additional constraints to provide obstacle avoidance [17], [18], [19]. As stated in [20], in the presence of pop-up obstacles, loss of recursive feasibility and undesirable aggressive tracking may occur, even for approaches designed for avoidance. In fact, depending on how the optimization problem is formulated, it is difficult to guarantee recursive feasibility in environments with a large number of obstacles since, due to obstructions, there may be no available path towards the goal and, also, considering the obstacles as constraints, the search-space for the optimization problem may become excessively small. Therefore, to work around that problem and still follow the idea of integrating control and planning strategies within a single Optimal Control Problem (OCP), analytical methods, such as artificial potential field-based methods [21], stand out due to a straightforward integration with the optimization problem by conveniently modifying only the value function. Some works considering artificial potential fields with MPC can be found in [22], [23], [24].

This work proposes a single-layer MPC strategy to provide motion systems with energy-awareness and obstacle avoidance capabilities. For that, the tracking MPC framework proposed in [25] is adapted by considering the artificial variables to not only avoid loss of recursive feasibility due to references changes but also to achieve obstacle avoidance feature with the help of artificial potential fields. Additionally, the rate capacity effect considered in [7] is used to obtain energy-awareness based on a simplified representation of the battery state-of-charge. Finally, to corroborate the proposed strategy, simulation results in a case study considering a quadrotor UAV are provided. Therefore, the contributions of this work can be summarized as: i) a novel energy-aware MPC strategy with obstacle avoidance feature, ii) the solution through an unified approach to prevent suboptimal responses and performance deterioration, and iii) the evaluation of the proposed strategy considering the quadrotor dynamic representation, unlike the kinematic one used in [7].

The remainder of this paper is organized as follows: Section II states the problem of solving the autonomous navigation while accounting for energy efficiency and providing obstacle avoidance features; Section III describes a simplified model for the battery state-of-charge dynamics based on the rate capacity effect; Section IV proposes the optimal control problem that handles in a single-layer the navigation, both planning and control, and energy-awareness; Section V presents the quadrotor UAV modeling, the control design parameters, and the numerical experiments used to corroborate the proposed control strategy; finally, Section VI concludes the work.

II. PROBLEM STATEMENT

Considering the battery rate capacity effect within a simplified model for its state-of-charge, as proposed in [7], the control problem in this work is the problem of designing an energy-aware controller able to perform autonomous navigation towards a given point with enhanced overall endurance

and with obstacle avoidance feature by means of an artificial reference obtained to avoid the obstacles.

For that, let the system dynamics be described by a finite-dimensional linear discrete-time system of the form

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k), \\ \mathbf{y}(k) &= \mathbf{C}\mathbf{x}(k). \end{aligned} \quad (1)$$

and let the state, input, and output vectors, respectively, $\mathbf{x}(k) \in \mathbb{R}^n$, $\mathbf{u}(k) \in \mathbb{R}^m$, and $\mathbf{y}(k) \in \mathbb{R}^p$, be constrained as

$$(\mathbf{x}(k), \mathbf{u}(k), \mathbf{y}(k)) \in \mathbb{X} \times \mathbb{U} \times \mathbb{Y}, \quad \forall k \in \mathbb{I}_{\geq 0}, \quad (2)$$

with $\mathbb{X}$, $\mathbb{U}$, and $\mathbb{Y}$ being, respectively, the set of admissible states, inputs, and outputs. Furthermore, $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times m}$, and $\mathbf{C} \in \mathbb{R}^{p \times n}$ are, respectively, the state, input, and output Jacobians linear model matrices. Typically, $\mathbf{y}(k)$, in equation (1), represents the motion system position, allowing the obstacle avoidance to be handled in the Euclidean space without loss of generality. Moreover, consider the following assumptions on the dynamical models.

Assumption 1. *The dynamical system (1) is controllable and the states are measured at each sampling time.*

Assumption 2. *The sets $\mathbb{X}$, $\mathbb{U}$, and $\mathbb{Y}$ are compact, and each set contains the origin.*

Furthermore, Problem 1 states the considered stabilization problem in terms of the motion system states.

Problem 1. *Consider a motion system described by (1) and subjected to the constraints (2) navigating in a cluttered environment with unknown obstacles. Furthermore, let $\mathbf{x}_r$ be an achievable desired state. The problem is to design a control law to make the error between the state and a constant reference arbitrarily small as time evolves while maximizing the battery overall state-of-charge and avoiding unknown obstacles, i.e., for a given $\epsilon > 0$ sufficiently small, as k goes to infinity, $\|\mathbf{x}(k) - \mathbf{x}_r\|_2 < \epsilon$.*

III. BATTERY STATE-OF-CHARGE DYNAMICAL MODELING

As proposed in [7], it is possible to account for the battery nonlinear behavior considering the factor known as rate capacity effect, which states that the available battery capacity decreases at a slower rate when lower discharging power is required. Thus, energy-aware controllers can be built in the understanding that by minimizing the battery discharging power, more energy can be extracted and enhanced endurance can be achieved for navigation.

Assumption 3. *The instantaneous discharging power drawn from the battery can be approximated by the Manhattan norm of the control inputs. In addition, the battery state-of-charge is measured at each sampling time.*

Let the battery state-of-charge at time-instant k be defined as

$$s(k) = \frac{Q(k)}{Q_{max}} 100, \quad (3)$$

with $Q(k)$ denoting the battery actual capacity, and Q_{max} denoting its nominal maximum capacity.

As presented in [7], the energy extraction from a battery and consequently, its state-of-charge, is subject to the rate capacity effect, which can be described by the Peukert's law

$$C_p = I^\beta T_d, \quad (4)$$

with C_p being the battery nominal capacity calibrated with one ampere discharging rate, I being the actual discharging current, T_d being the time to completely discharge the battery, and $\beta > 1$ being the Peukert constant. Since for a given battery the nominal capacity is constant, following the idea of an exponential relation between the discharge current and the discharging time, the state-of-charge dynamics can be described as [7]

$$s(k+1) = s(k) - \mu \|\mathbf{u}(k)\|_1^\beta, \quad (5)$$

where $\|\mathbf{u}(k)\|_1$ denotes the control input's 1-norm, and

$$\mu = \frac{T_s \delta}{3600 Q_{max}}, \quad (6)$$

with T_s being the sampling period in seconds, $\delta \geq 1$ being the discharge efficiency coefficient, and Q_{max} being given in Watts per hour.

Aiming to avoid the battery overuse and to prevent life-damaging effects, the state-of-charge and the discharging power can be constrained to a specific range as

$$s(k) \in [s_{min}, s_{max}], \quad (7)$$

$$\|\mathbf{u}(k)\|_1 \in [0, u_{max}], \quad (8)$$

with s_{min} and s_{max} denoting the minimum and the maximum state-of-charge, and u_{max} denoting the maximum discharging power.

Furthermore, since the unlimited instantaneous change of the discharging power are not realistic due to the battery dynamics, an additional constraint is imposed on the control input variation to indirectly bound the discharging power between samples

$$|\mathbf{u}(k+1) - \mathbf{u}(k)| \leq \delta u_{max}, \quad (9)$$

with δu_{max} being the maximum step between samples.

IV. OPTIMAL CONTROL PROBLEM

Considering the position stabilization control problem for motion systems, it is possible to propose a single-layer finite-horizon OPC addressing the trade-off between the tracking performance and the energy consumption while accounting for the obstacle avoidance. For that, let $N \in \mathbb{I}_{>0}$ be the horizon length and $\mathbf{x}_a(k) \in \mathbb{R}^n$, $\mathbf{u}_a(k) \in \mathbb{R}^m$, and $\mathbf{y}_a(k) \in \mathbb{R}^p$ be the artificial variables associated, respectively, with the state, input, and output vectors. Moreover, let $(\cdot)_{\underline{\gamma}}$ denotes a finite sequence composed of i elements evaluated at the integers set $\mathbb{I}_{0:i}$, i.e., integers going from 0 to i . For instance, $\underline{\mathbf{x}} = \{\mathbf{x}(k), \mathbf{x}(k+1), \dots, \mathbf{x}(k+i)\}$. Finally, $\mathbf{y}_o^\ell$ denotes the point on the surface of the ℓ -th obstacle having the smallest distance from the motion system position.

Definition 1. (Open-loop OCP for energy-aware control) Given a horizon N and the optimization parameters $\mathbf{z}_1 \in \mathbb{R}^n$, $z_2 \in \mathbb{R}$, $\mathbf{y}_r \in \mathbb{R}^n$, and $\mathbf{y}_o^\ell \in \mathbb{R}^p$, respectively, the measured state, the measured state-of-charge, the desired reference, and the ℓ -th obstacle position, the open-loop optimal

control problem consists in finding the optimal sequence for the decision variables $\underline{\mathbf{x}}$, $\underline{\mathbf{u}}$, $\mathbf{x}_a$, $\mathbf{u}_a$, $\mathbf{y}_a$, and $\underline{s}$, solving

$$\begin{aligned} V_N^O(\mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell) = & \min_{\underline{\mathbf{x}}, \underline{\mathbf{u}}, \mathbf{x}_a, \mathbf{u}_a, \mathbf{y}_a, \underline{s}} V_N(\mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell; \underline{\mathbf{x}}, \underline{\mathbf{u}}, \mathbf{x}_a, \mathbf{y}_a) \\ \text{s.t. } & \mathbf{x}(k) = \mathbf{z}_1, \quad (10a) \\ & s(k) = z_2, \quad (10b) \\ & \mathbf{x}(k+j+1) = \mathbf{A}\mathbf{x}(k+j) + \mathbf{B}\mathbf{u}(k+j), \quad j \in \mathbb{I}_{0:N-1}, \quad (10c) \\ & (\mathbf{x}(k+j), \mathbf{u}(k+j)) \in \mathbb{X} \times \mathbb{U}, \quad j \in \mathbb{I}_{0:N}, \quad (10d) \\ & s(k+j+1) = s(k+j) - \mu \|\mathbf{u}(k+j)\|_1^\beta, \quad j \in \mathbb{I}_{0:N-1}, \quad (10e) \\ & s(k+j) \in [s_{min}, s_{max}], \quad j \in \mathbb{I}_{0:N}, \quad (10f) \\ & \|\mathbf{u}(k+j)\|_1 \in [0, u_{max}], \quad j \in \mathbb{I}_{0:N}, \quad (10g) \\ & |\mathbf{u}(k+j+1) - \mathbf{u}(k+j)| \leq \delta u_{max}, \quad j \in \mathbb{I}_{0:N-1}, \quad (10h) \\ & (\mathbf{x}_a(k), \mathbf{u}_a(k)) \in \mathbb{X} \times \mathbb{U}, \quad (10i) \\ & \mathbf{x}_a(k) = \mathbf{A}\mathbf{x}_a(k) + \mathbf{B}\mathbf{u}_a(k), \quad (10j) \\ & \mathbf{y}_a(k) = \mathbf{C}\mathbf{x}_a(k), \quad (10k) \\ & \mathbf{x}(k+N) = \mathbf{x}_a(k). \quad (10l) \end{aligned}$$

An optimal solution for the open-loop optimal control problem (10) gives the optimal control sequence $\underline{\mathbf{u}}^O(\mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell) = \{\mathbf{u}^O(k; \mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell), \dots, \mathbf{u}^O(k+N-1; \mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell)\}$. Thus, considering the receding policy of MPC controllers, the optimal control law for the motion system at each sampling period is given by $\mathbf{K}_N(k; \mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell) = \mathbf{u}^O(k; \mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell)$. Further, the cost functional of (10) is formulated to combine the different objectives of providing motion control, energy-awareness, and obstacle avoidance at the same time. For that, we can split the value function of (10) into a stabilizing MPC stage cost accounting for the discharge rate, $\ell_s(\cdot)$, and a cost for avoidance, $\ell_o(\cdot)$, yielding to

$$\begin{aligned} V_N(\mathbf{z}_1, z_2, \mathbf{y}_r, \mathbf{y}_o^\ell; \underline{\mathbf{x}}, \underline{\mathbf{u}}, \mathbf{x}_a, \mathbf{y}_a) = & \sum_{j=0}^{N-1} \ell_s(\mathbf{x}(k+j), \mathbf{x}_a(k), \mathbf{u}(k+j)) + \ell_o(\mathbf{y}_a(k), \mathbf{y}_r, \mathbf{y}_o^\ell). \quad (11) \end{aligned}$$

Aiming at motion control and energy-awareness, the stabilizing stage cost can be defined as a slightly modified version of the standard MPC quadratic cost by directly minimizing the discharging power, yielding to

$$\begin{aligned} \ell_s(\mathbf{x}(k+j), \mathbf{x}_a(k), \mathbf{u}(k+j)) = & \|\mathbf{x}(k+j) - \mathbf{x}_a(k)\|_{\mathbf{Q}}^2 + r \|\mathbf{u}(k+j)\|_1^2, \quad (12) \end{aligned}$$

with $j \in \mathbb{I}_{0:N-1}$, $\mathbf{Q} > 0$ being a weight matrix, and $r > 0$ being a weight constant. It is worth noticing that the stabilizing stage cost is defined with respect to the artificial state instead of the reference one, which, together with the terminal constraint defined with respect to the artificial variable, increases the controller domain of attraction. Furthermore, since an equality terminal constraint was defined in (10l), the stabilizing stage cost does not require terminal cost in its formulation.

Aiming to reduce the computational complexity of the one-norm minimization problem, the well-known equivalence between linear programming and the basis pursuit problem can be used [26]. For that, let $\underline{\gamma} \in \mathbb{R}^{m \times N}$ be a new decision variable of the optimization problem (10) in such a way that the stabilizing stage cost (12) can be rewritten as

$$\begin{aligned} \ell_s(\mathbf{x}(k+j), \mathbf{x}_a(k), \mathbf{u}(k+j)) = & \|\mathbf{x}(k+j) - \mathbf{x}_a(k)\|_{\mathbf{Q}}^2 + r \|\underline{\gamma}(k+j)\|^2. \quad (13) \end{aligned}$$

Moreover, the additional constraints must be imposed to the optimization problem

$$\mathbf{u}(k+j) - \gamma(k+j) \leq \mathbf{0}, \quad j \in \mathbb{I}_{0:N-1}, \quad (14)$$

$$-\mathbf{u}(k+j) - \gamma(k+j) \leq \mathbf{0}, \quad j \in \mathbb{I}_{0:N-1}, \quad (15)$$

with $\mathbf{0}$ being a vector of zeros with appropriate dimension.

As for the obstacle avoidance, the cost $\ell_o(\cdot)$ is defined as a combination of attraction and repulsive potential fields, allowing it to be decomposed as

$$\ell_o(\mathbf{y}_a(k), \mathbf{y}_r, \mathbf{y}_o^\ell) = \ell_{at}(\mathbf{y}_a(k), \mathbf{y}_r) + \ell_{rp}(\mathbf{y}_a(k), \mathbf{y}_o^\ell), \quad (16)$$

where $\ell_{at}(\cdot)$ denotes an attraction potential cost and $\ell_{rp}(\cdot)$ denotes a repulsive potential cost.

On one hand, the attraction field cost, which is a function that increases with the distance between the artificial output and the desired reference, can be written as a quadratic function, yielding to

$$\ell_{at}(\mathbf{y}_a(k), \mathbf{y}_r) = \|\mathbf{y}_a(k) - \mathbf{y}_r\|_{\kappa}^2, \quad (17)$$

where $\kappa > 0$ is a weighting matrix.

On the other hand, repulsive potential cost, which is a function inversely proportional to the distance between the artificial output and the ℓ -th obstacle position, can be written as a logistic function, as proposed in [24], which yields to

$$\ell_{rp}(\mathbf{y}_a(k), \mathbf{y}_o^\ell) = \frac{\lambda}{1 + e^{p_1(\|\mathbf{y}_a(k) - \mathbf{y}_o^\ell\|_{\Gamma}^2 - p_2)}}, \quad (18)$$

with $\lambda > 0$, $p_1 > 0$, $p_2 > 0$ being adjustment parameters and $\Gamma > 0$ being a weighting matrix.

In the optimal control problem of Definition 1, the constraints (10a) and (10b) define initial conditions, respectively, for the states and the battery state-of-charge. Further, the prediction process for the motion system and the battery usage are subjected to the dynamical models (1) and (5) with the constraints (10c) and (10e). Since the motion system is constrained to admissible state and input sets, the constraints (10d) and (10i) are considered and, to account for the battery operation, (10f), (10g), and (10h) are imposed as defined in equations (7) to (9). Finally, constraints (10j) and (10k) ensure that the artificial variable is an equilibrium point reachable in N steps, which is enforced by the terminal equality constraint (10l). Furthermore, to achieve the terminal equality condition successfully, a controllability condition is required to ensure the existence of a control sequence that can transfer the system from any $\mathbf{x}(k)$ to $\mathbf{x}_a(k)$ in N steps.

Assumption 4. *The prediction horizon N is such that the N -controllability matrix of a system $(\mathbf{A}, \mathbf{B})$, $\mathbf{Co}_N = [\mathbf{A}^{N-1}\mathbf{B} \ \dots \ \mathbf{A}\mathbf{B} \ \mathbf{B}]$, has full rank.*

It is worth highlighting two points regarding the OCP constraints. First, the output reference does not appear in the constraints, thus making the OCP feasible for any changing reference. Second, the terminal equality constraint is not a tough one in this formulation, since the presence of the artificial variables relaxes it. Therefore, the resulting controller should present enlarged regions of attraction when compared to standard stabilizing MPC schemes.

V. NUMERICAL EXAMPLE: A QUADROTOR UAV

This section presents simulation results obtained to corroborate the effectiveness of the proposed energy-aware MPC for avoidance to provide safe navigation in cluttered environments while maximizing the battery state-of-charge. For the simulations, a quadrotor UAV is considered.

A. Quadrotor UAV Modeling

For the sake of completeness, this subsection briefly describes the dynamic equations of motion for a quadrotor UAV. Aiming to better emulate the vehicle dynamics during the simulation, a more complete model is presented taking into account the coupling between translational and rotational dynamics due to the displacement between the quadrotor's geometric center and its center of mass [27]. However, for control design purposes, this displacement is neglected and a simplified version of the model is considered [28].

The quadrotor can be seen as a rigid body described by a frame $\mathcal{B}$ rigidly attached to its center of rotation, a frame $\mathcal{C}$ rigidly attached to its center of mass, and an inertial frame $\mathcal{I}$ (see Fig. 1). The position of the body frame's origin represented in $\mathcal{I}$ is given by $\boldsymbol{\xi} = [x \ y \ z]'$ and its attitude by $\boldsymbol{\eta} = [\phi \ \theta \ \psi]'$, considering the Euler angles with the ZYX convention about local axes. Fig. 1 shows the four propellers tilted towards the origin of $\mathcal{B}$ by an angle α , which is introduced, as explained in [28], to increase the controllability by allowing direct actuation on x and y without the necessity to employ augmented state vector nor cascade control strategies. On the other hand, it is worth noting that this procedure reduces the total lift force in the body Z-axis direction, thus, α is expected to be small aiming at a good trade-off between controllability and lifting power. Moreover, for modeling purposes, since the quadrotor is assumed to be a rigid body, then, the distance r between the frames $\mathcal{B}$ and $\mathcal{C}$ is constant. Thus, the generalized coordinates describing the quadrotor UAV motion can be chosen as $\mathbf{q} = [\boldsymbol{\xi}' \ \boldsymbol{\eta}']'$. Finally, the system's inputs are $\mathbf{u} = [f_1 \ f_2 \ f_3 \ f_4]'$, with f_i being the thrust force generated by the i -th rotor.

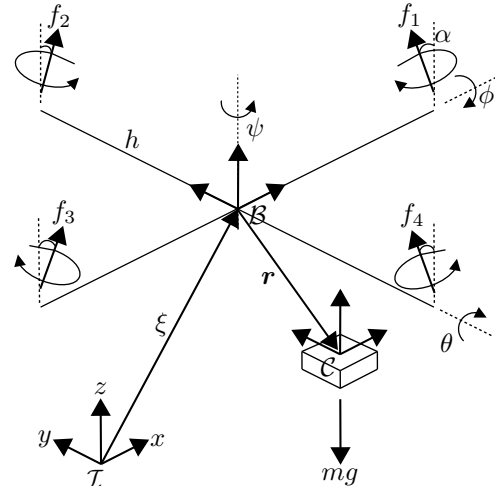


Fig. 1. Quadrotor UAV frames and variables definition.

The dynamic model of the quadrotor UAV can be described by the Euler-Lagrange equations of motion as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = B(q)u, \quad (19)$$

with $M(q) \in \mathbb{R}^{n \times n}$ denoting the inertia matrix, $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$ denoting the Coriolis and centripetal forces matrix, $G(q) \in \mathbb{R}^n$ denoting the gravitational force vector, and $B(q) \in \mathbb{R}^{n \times m}$ denoting the force matrix.

The inertia matrix and the gravitational force vector can be written, respectively, as

$$M(q) = \begin{bmatrix} m\mathbb{1} & -mR_B^T S(r) W_\eta \\ -m(R_B^T S(r) W_\eta)' & J(\eta) \end{bmatrix}, \quad (20)$$

$$G(q) = \begin{bmatrix} 0 \\ 0 \\ mg \\ -mg(-r_y c_\theta c_\phi + r_z s_\theta s_\phi) \\ -mg(r_x c_\theta + r_y s_\theta s_\phi + r_z s_\theta c_\phi) \\ 0 \end{bmatrix}, \quad (21)$$

where R_B^T denotes the rotation matrix between $\mathcal{B}$ and $\mathcal{I}$, $S(\cdot)$ denotes the skew symmetric matrix [29], $r = [r_x \ r_y \ r_z]'$, m is the UAV mass, g is the gravitational acceleration, $\mathbb{1}$ represents an identity matrix with appropriate dimension, $s(\cdot) = \sin(\cdot)$, and $c(\cdot) = \cos(\cdot)$. Further, $J(\eta) = W_\eta' I W_\eta$, with I being the moment of inertia and

$$W_\eta = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix}.$$

Additionally, the Coriolis and centripetal forces matrix, $C(q, \dot{q})$, is obtained by using the Christoffel symbols of first kind [29].

The generalized force vector can be defined as

$$\begin{bmatrix} f_\xi \\ \tau_\eta \end{bmatrix} = B(q)u, \quad (22)$$

where $f_\xi = R_B^T f_a$ represents the translational force vector, and $\tau_\eta = W_\eta' \tau_a$ represents the total attitude moments, both expressed in the inertial reference frame.

Inspecting Fig. 1, it is easy to see that f_a and τ_a can be expressed as

$$f_a = \begin{bmatrix} s_\alpha (f_3 - f_1) \\ s_\alpha (f_4 - f_2) \\ \left(\sum_{i=1}^4 c_\alpha f_i \right) \end{bmatrix}, \quad \tau_a = \begin{bmatrix} (f_2 - f_4) h c_\alpha \\ (f_3 - f_1) h c_\alpha \\ \sum_{i=1}^4 \tau_{M_i} c_\alpha \end{bmatrix}, \quad (23)$$

where h is the distance between the rotors and the center of rotation, and τ_{M_i} is the motor torsion effort.

The lift force generated by each rotor can be approximated by $f_i = b\Omega_i^2$ and its torsion effort by $\tau_{M_i} = k_\tau \Omega_i^2$, with Ω_i being the angular velocity of the i -th rotor around its axis, b being the thrust coefficient, and $k_\tau > 0$ being a drag coefficient. Therefore, using equations (22) and (23), the force matrix $B(q)$ can be written as

$$B(q) = \begin{bmatrix} R_B^T & 0 \\ 0 & W_\eta' \end{bmatrix} \begin{bmatrix} -s_\alpha & 0 & s_\alpha & 0 \\ 0 & -s_\alpha & 0 & s_\alpha \\ c_\alpha & c_\alpha & c_\alpha & c_\alpha \\ 0 & h c_\alpha & 0 & -h c_\alpha \\ -h c_\alpha & 0 & h c_\alpha & 0 \\ \frac{k_\tau}{b} c_\alpha & -\frac{k_\tau}{b} c_\alpha & \frac{k_\tau}{b} c_\alpha & -\frac{k_\tau}{b} c_\alpha \end{bmatrix}. \quad (24)$$

After all terms of the equation (19) are obtained, it is possible to rewrite the equations of motion in the state-space representation, yielding to

$$\dot{x} = f(x, u) = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} M^{-1} [Bu - C\dot{q} - G] \end{bmatrix}. \quad (25)$$

Additionally, with the purpose to apply the proposed energy-aware MPC strategy, the model (25) is simplified by disregarding the coupling between the translational and rotational dynamics and, thereafter, it is linearized and discretized. Therefore, by linearizing the system around an equilibrium point, it yields to

$$\Delta \dot{x} = A \Delta x + B \Delta u, \quad (26)$$

where

$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{\substack{x=x^{eq} \\ u=u^{eq}}}, \quad B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{\substack{x=x^{eq} \\ u=u^{eq}}}, \quad (27)$$

with x^{eq} and u^{eq} being the equilibrium states and inputs obtained considering $q^{eq} = [x^{eq} \ y^{eq} \ z^{eq} \ \phi^{eq} \ \theta^{eq} \ \psi^{eq}]'$ and $u^{eq} = [f_1^{eq} \ f_2^{eq} \ f_3^{eq} \ f_4^{eq}]'$, with f_i^{eq} being the thrust required to keep the vehicle in hovering without external disturbances.

Since x , y , and z are external variables for the quadrotor UAV dynamic model (19) (i.e. they do not shape the inertia matrix and, consequently, the Coriolis matrix), the equilibrium condition q^{eq} only enforces the UAV orientation in hovering with null translational and angular velocities. As for the position (x^{eq}, y^{eq}, z^{eq}) , any point in the robot workspace can be used. Furthermore, due to the equilibrium condition on the artificial trajectory, constraints (10j) and (10k), the controller will be constantly keeping the system close to the linearization condition.

Finally, the discrete linearized model can be obtained after mapping (26) from the continuous-time to the discrete-time domain, yielding to

$$\Delta x(k+1) = A_d \Delta x(k) + B_d \Delta u(k), \quad (28)$$

being A_d and B_d obtained after discretizing the model using a fixed-step fourth-order Runge-Kutta integrator with sampling time T_s .

B. Numerical Results

Aiming to assess the features of the proposed control strategy, the quadrotor UAV workspace is obstructed by 9 rectangle-shaped obstacles, and a 3D lidar-like sensor is emulated to detect them within a spherical range of 3 meters based on the robot global position and the environment map. For that, the map was split into grids and, from the UAV position, the distance to the detected obstacles was calculated. However, it should be highlighted that the map knowledge is only required for the detection algorithm since from the control algorithm standpoint there is no assumption being made about the knowledge of the obstacles' position or their shapes. Besides, inspired by the parameters presented in [7], we considered a battery that takes around 30 minutes to fully discharge under a constant discharging power of 36 W. For that, we choose $\delta = 18$, $Q_{max} = 36$, $\beta = 5/3$, and $s(0) = 80$. Additionally, the battery is required to operate between $s_{min} = 20$ and $s_{max} = 90$ of state-of-charge with

a maximum allowed discharging power of $u_{max} = 30$ and, to constraint the discharging power between samples, the control input variation is limited to $\delta u_{max} = 0.5$. Moreover, the quadrotor UAV parameters are presented in Table I and, for linearization purposes, the sampling time considered is $T_s = 0.01$ seconds with the equilibrium conditions $\phi^{eq} = \theta^{eq} = \psi^{eq} = 0$, $f_1^{eq} = f_2^{eq} = f_3^{eq} = f_4^{eq} = 5.51$.

As for the controller parameters, the horizon length is chosen as $N = 50$, and the weighting variables for the stabilizing stage cost are the diagonal matrix $\mathbf{Q} = \text{diag}(0.25, 0.25, 0.25, 0.40, 0.40, 0.10, 0.25, 0.25, 0.25, 0.01, 0.01, 0.02)$ and the constant $r = 1$. Besides, the parameters for the potential field-based cost are $\kappa = \text{diag}(1000, 1000, 1000)$, $\lambda = 50000$, $p_1 = 5$, $p_2 = 2$, and $\mathbf{\Gamma} = \text{diag}(1, 1, 0.1)$. Furthermore, the admissible inputs set is $\mathbb{U} = \{(f_1, f_2, f_3, f_4) \in \mathbb{R}^4 \mid 0 \leq f_1, f_2, f_3, f_4 \leq 12\}$, and the admissible states set is defined as $\mathbb{X} = \{x \in \mathbb{R}^{12} \mid x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}, |\phi| < \pi/2, |\theta| < \pi/2, |\psi| < \pi/6, |\dot{x}| < 5, |\dot{y}| < 5, |\dot{z}| < 5, |\dot{\phi}| < \pi/2, |\dot{\theta}| < \pi/2, |\dot{\psi}| < \pi/2\}$. For comparison purposes, simulations are also carried out considering the MPC controller proposed without the energy-awareness, i.e., without the constraints (10e), (10f), (10g), and replacing the discharging power term of (12) by the standard squared weighted two norm of the error between $\mathbf{u}$ and $\mathbf{u}_a$. Aiming to obtain a comparable cost functional between the strategies, the parameter considered to weight the two norm is 4 times bigger than the one used for the one norm. Thus, with the exception of the weight matrix $\mathbf{R} = \text{diag}(4, 4, 4, 4)$, the same parameters defined for the energy-aware controller are considered. Finally, the simulations are performed with MATLAB® using the CasADI Toolbox [30] with the IPOPT solver [31].

TABLE I
QUADROTOR UAV MODEL PARAMETERS.

Parameter	Value
m	2.24 [kg]
h	0.332 [m]
g	9.81 [m/s ²]
b	$9.5 \cdot 10^{-6}$ [Ns ²]
k_τ	$1.7 \cdot 10^{-7}$ [Nms ²]
α	5 [°]
I_{xx}	0.0363 [kgm ²]
I_{yy}	0.0363 [kgm ²]
I_{zz}	0.0615 [kgm ²]
r_x	-0.00069 [m]
r_y	-0.0014 [m]
r_z	-0.0311 [m]

In this simulation scenario, the aerial robot is required to perform a task autonomously while maximizing the state-of-charge of the battery and avoiding the obstacles. Therefore, starting from its initial position $(-21, -12, 0)$ in the lower-right side, see Fig. 2, it navigates towards the goal $(18.75, 6, 15)$, which is depicted as a black dot in the upper-left side. When the distance between the measured position and the goal is smaller than 20 centimeters, the task is assumed completed and the simulation is stopped. In Fig. 2,

it is possible to verify that the robot was able to perform the task even in the presence of unknown obstacles obstructing any obvious path to the target, which validates the ability to find a path around the obstacles that appear during the navigation with the help of the artificial variables and the potential artificial fields. Besides, this figure also shows that the paths taken by the robots controlled by the strategies with energy-awareness, blue path, and without energy-awareness, red path, are similar. As for the time spend to execute the task, the energy-aware controller took 117.00 seconds, while the avoidance controller took 129.56 seconds. It is worth noting that the maximization of the state-of-charge for aerial vehicles cannot be seen as a trade-off between the control effort and the time to accomplish the task, once keeping the UAV hovering already spends an important amount of energy. Therefore, since only seeking for smaller accelerations may result in a slow system drawing more energy from the battery, the energy-aware approach will consist of finding paths with smaller discharging rates while satisfying the battery's operational constraints to prevent any life-damaging effects.

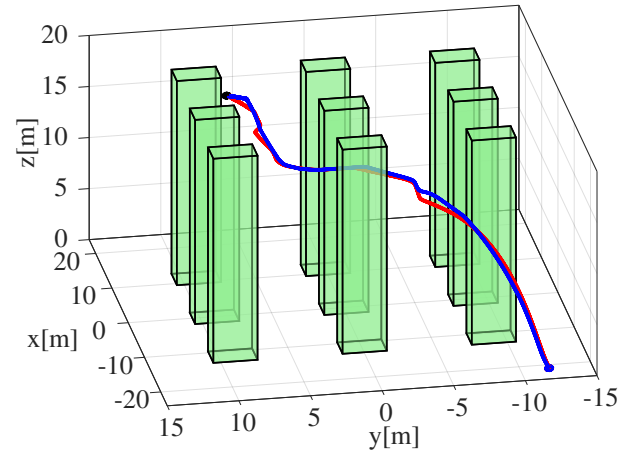


Fig. 2. Trajectory performed by the UAV controlled with energy-awareness, blue line, and without energy-awareness, red line, to safely execute the task.

Fig. 3 shows the time evolution of the quadrotor UAV orientation for both controllers. Since at the end of the task execution the UAV must reach an equilibrium, it is expected that the roll and pitch angles will stably converge to zero while the yaw angle will converge to a constant value, which can be observed in Fig. 3. Similarly, despite some oscillations, the linear and angular velocities will also stably converge to zero at the end of the task execution, which is shown in Fig. 4 and Fig. 5. Particularly, the oscillation on $\dot{y}$ and $\dot{\phi}$ around 40 and 60 seconds explains the roll motion, ϕ , observed in Fig. 3 for the quadrotor UAV controlled without energy-awareness. This situation occurred because the obstacle was detected at the limit of the sensor's range, which created a on/off behavior on the repulsive field contribution to the overall cost functional. Furthermore, Fig. 6 shows the lift forces applied to the quadrotor UAV, which were kept stable throughout the navigation despite some oscillations when a sudden change of direction is required to

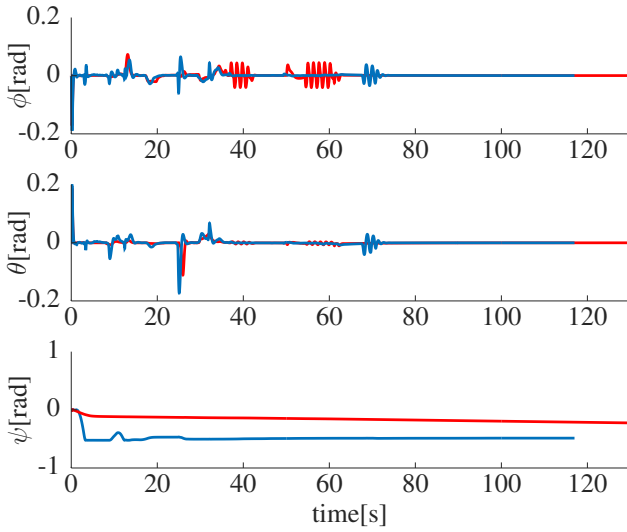


Fig. 3. Time evolution of the quadrotor UAV orientation with energy-awareness, blue line, and without energy-awareness, red line.

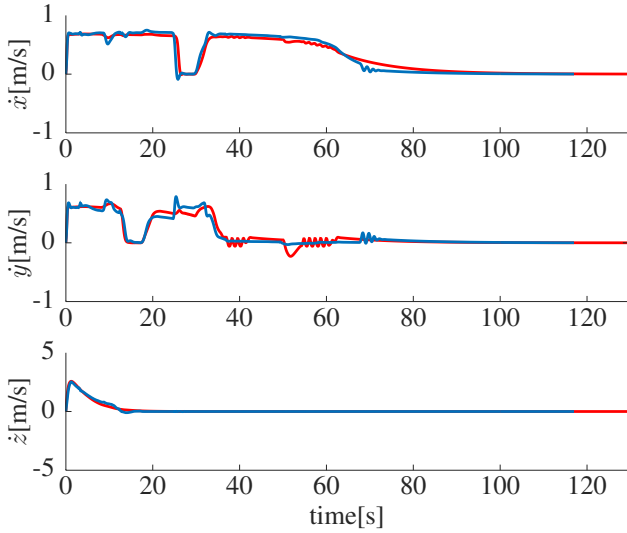


Fig. 4. Time evolution of the quadrotor UAV linear velocity with energy-awareness, blue line, and without energy-awareness, red line.

avoid a new detected obstacle. Finally, despite the challenges of handling different goals in a single-layered controller, for the considered simulation scenario, it is possible to state that the energy-aware model predictive control strategy for avoidance was able to ensure convergence and closed-loop stability for the problem addressed.

Fig. 7 shows the discharging power, which is assumed to be approximated by the Manhattan norm of the control inputs. It is possible to see that the energy-aware controller, blue line, presents smaller discharging power, mainly when maneuvers that demand more energy are performed. For instance, at the beginning of the task execution, the pure avoidance controller is more active in achieving the desired altitude. This results in higher control effort and a more steep climb to the desired altitude. As consequence, the discharging power at the beginning of the simulation is greater for the controller without energy-awareness, as expected.

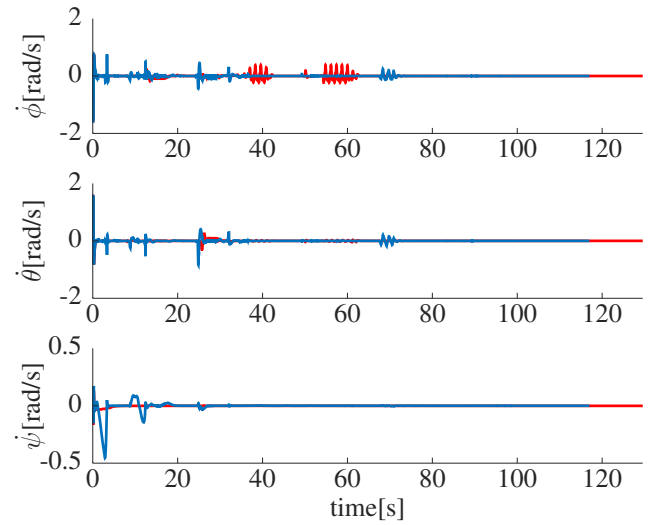


Fig. 5. Time evolution of the quadrotor UAV angular velocity with energy-awareness, blue line, and without energy-awareness, red line.

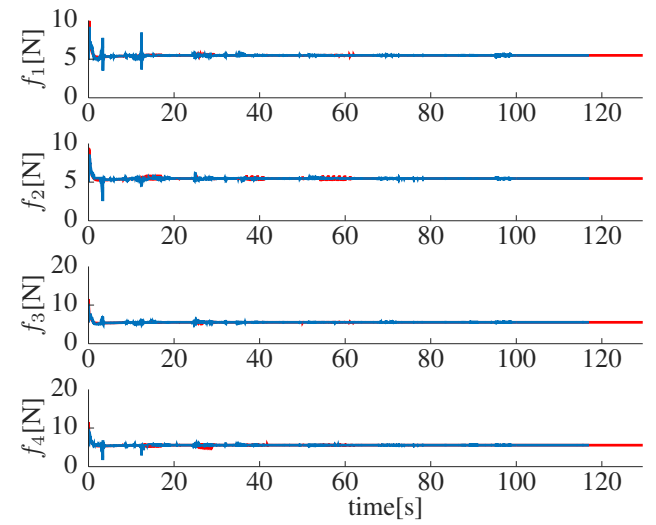


Fig. 6. Applied lift forces to the UAV controlled with energy-awareness, blue line, and without energy-awareness, red line.

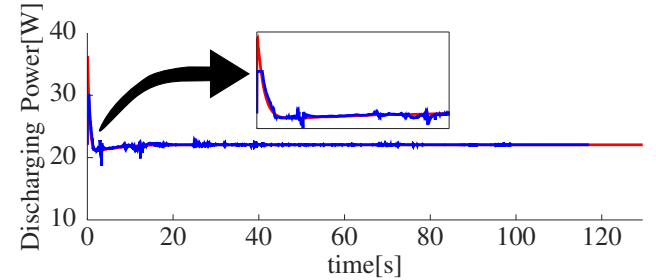


Fig. 7. Time evolution of the discharging power considering the energy-aware controller, blue line, and the avoidance only controller, red line.

As for the state-of-charge, the proposed energy-aware strategy finishes the task with a level of 77.17 % while the controller without energy-awareness ends with 76.88 %. Therefore, it can be said that, in relative terms, the energy-aware controller was able to perform the task drawing only 90.30 % of the energy that was drawn by the avoid-

ance controller. Besides, when looking at the total energy composed of kinetic and potential energy, the energy-aware controller ends up with a power of 31.69 kW, while the controller without energy-awareness ends with 31.76 kW. It is worth noting that any controller can be tuned by the designer to achieve energy efficiency once it is known what energy consumption situations must be avoided. However, the proposed strategy provides the designer with additional parameters to directly enhance energy efficiency. Further, most of the energy consumption reduction in the presented simulation was obtained during the maneuvers that required lifts, such as the climb at the beginning of the simulation. Thus, in scenarios where high control levels are required, e.g. aggressive maneuvers, the relative energy consumption reduction should be greater. Finally, by considering the operational constraints (10f), (10g), and (10h) explicitly, the life span of the battery can be extended when compared to other control strategies that do not consider energy-awareness in their formulations, e.g., standard stabilizing MPC schemes.

VI. CONCLUSIONS

In this work, the energy efficiency problem was addressed considering the nonlinear dynamics of the battery through its rate capacity effect. For that, a single-layer model predictive control strategy able to perform obstacle avoidance during autonomous navigation in cluttered environments while accounting for energy efficiency was proposed. The obstacle avoidance feature was obtained by combining the ideas of artificial navigation fields within the tracking MPC framework through suitable artificial variables. On the other hand, the energy-awareness is obtained considering the rate capacity effect of electric batteries to limit the discharging power during navigation aiming to maximize the overall state-of-charge. To corroborate the proposed strategy, numerical results in a case study considering a quadrotor UAV were provided.

The results obtained were based on a simplified dynamic description of the battery and on the strong assumption that the Manhattan norm could describe well the discharging power required, which is not true in most cases. Besides, due to the model limitation, it was not possible to exploit some operational conditions and constraints of the battery that directly affects its state-of-health and, ultimately, its lifespan. Therefore, in future works, such assumptions will be disregarded, and the problem will be addressed considering more representative models for the battery. Furthermore, the obtained gain in the state-of-charge maximization ended-up being small because the resulting artificial trajectory required only low acceleration maneuvers that did not need much power for being performed. Two reasons explain this behavior, the first one is related to the limitations of the linearized model, mainly, to describe the orientation dynamics far from the linearizing condition; the second one, on the other hand, is related to the equilibrium condition for the artificial trajectory that requires the system to be with zero velocity in N steps, which clearly limits the maximum acceleration of the closed-loop system. Thus, in future works, the pro-

posed control strategy will be extended to nonlinear systems. Also, solutions to drop the equilibrium condition for the artificial reference without losing the stability and feasibility properties that come from the tracking MPC formulations will be proposed. The main goal is to be able to evaluate the system during aggressive maneuvers where conflicting goals such as minimum time and energy efficiency will be present. This work was mostly concerned with providing a proof of concept regarding the possibility to design a novel single-layer control strategy able to provide the closed-loop system with energy-awareness and obstacle avoidance capabilities. However, to achieve experimental results, the overall computational time to solve the optimization problem still needs to be reduced. Therefore, in future work, strategies to solve the optimization problem within the system sampling time will be investigated. Finally, future work will also provide formal analysis of the proposed control strategy stability properties.

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